

Tirth Shah

Milpitas, California, United States | shahtirth126@gmail.com | (530) 321-6233 | github.com/shahtirth07 | linkedin.com/in/shahtirth07 | tirthshah.com

PROFESSIONAL SUMMARY

Backend engineer and AI systems architect with 3+ years building scalable services and agentic AI systems. Expertise in distributed systems (Go, Python, Node.js), multi-agent orchestration, and production infrastructure. Published researcher on conversational AI and NLP systems.

WORK EXPERIENCE

California State University, Chico | Research Assistant Sep 2024 -

- Built RAG pipelines with grounding and evaluation frameworks for conversational AI, reducing hallucination rate by 40% on noisy real-world inputs
- Co-authored 4 peer-reviewed publications on conversational agents and intelligent systems; research presented at ASEE Zone 4 Conference

SAP UCC Chico State | Forward Deployed Software Engineer May 2024 - May 2026

- Architected and shipped a RAG-based agent for product discovery across 30+ institutional partners, embedding 50K+ catalog items with semantic search; improved query relevance by 35%
- Engineered real-time notification pipeline using WebSockets, Redis pub/sub, and event-driven architecture; reduced latency from 2+ hours to under 500ms, serving 10K+ concurrent users
- Owned full-stack development of React + Node.js microservices platform; coordinated with 30+ institutional stakeholders in bi-weekly requirements sessions to shape roadmap and ship production features within sprints

Onix (Formerly Datametica Pvt Ltd) | Associate Member of Technical Staff Oct 2021 - Jul 2024

- Architected TDD-driven data migration framework supporting DataStage, Informatica, Alteryx; achieved 95%+ test coverage and improved downstream compatibility by 60%
- Built scalable synthetic data generation pipeline in Groovy/Scala; generated 1M+ records per run, cutting QA cycle time by 30 hours/month and enabling parallel test execution
- Designed and maintained multi-team CI/CD infrastructure using GitHub Actions and Jenkins; improved deployment frequency by 25% and reduced production incident recovery time by 40%
- Led monolith-to-microservices migration using Spring Boot, PostgreSQL, React; reduced API latency by 80% and enabled independent service scaling

PROJECTS

MockMouse: Agentic Browser Automation Framework

- Architected multi-agent orchestration system using Claude LLM for autonomous task planning and execution; converts natural language into sandboxed browser actions with tool execution isolation
- Engineered distributed agent runtime with Next.js frontend, FastAPI backend, WebSockets for real-time messaging, and Prisma for state persistence; handle 10K+ concurrent task requests
- Achieved Top 3 finish in Dev Tools category at Cal Hacks 2025 (9th overall)

Déjà Query: Token-Efficient NL-to-SQL Agent

- Built dual-mode query execution system: cold path compiles natural language + schema to SQL (stores as reusable template), warm path recognizes repeat queries and slot-fills at zero LLM tokens, achieving 95% token reuse on repeated question patterns
- Implemented semantic similarity matching using EverOS embeddings with MiniLM fallback; guarded by confidence thresholds and lexical safety rules to prevent template misapplication
- Deployed cost tracking and analytics dashboard; enables live visibility into per-pattern token spend and cumulative savings across query workload

Signal Room: Multi-Agent Lead Response System

- Designed multi-agent graph-based memory system (FalkorDB) with live web retrieval for sales lead qualification; human-in-the-loop approval gate ensures business logic safety
- Orchestrated agent coordination across FastAPI and React services using Apache Iggy event streaming; enables scalable, asynchronous task processing
- Reduced manual lead triage time by 60% while maintaining high-quality response accuracy

ChatBook: Retrieval-Augmented Learning System

- Designed RAG pipeline for semantic document search using LLM embeddings; enables students to query uploaded course materials conversationally
- Built full-stack system with React frontend, Flask backend, MongoDB Atlas vector store; deployed and iterated based on 200+ user sessions

BookMatch: Hybrid Recommendation Engine

- Built hybrid recommender system over 25K book catalog: collaborative filtering (scikit-learn KNN) for users with history, BERT embeddings for cold-start
- Deployed embeddings via Elasticsearch with cosine similarity; validated accuracy against 200+ manually-tagged recommendations

EDUCATION

California State University, Chico Master of Science, <i>Computer Science</i>	Aug 2024 - Jul 2026 GPA: 3.96
Savitribai Phule Pune University Bachelor of Engineering, <i>Electronics and Telecommunication</i>	Aug 2018 - May 2022 GPA: 3.74

SKILLS

Programming Languages: Python, Java, Go, Swift, Groovy, Kotlin, Scala, JavaScript, TypeScript

Frameworks & Libraries: React, Node.js, FastAPI, GraphQL, Spring Boot, Flask, LangChain, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Databases: PostgreSQL, MySQL, MongoDB, Redis, MongoDB Atlas, Elasticsearch, Prisma

Tools & Software: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Prometheus, Grafana, Git, browseruse, Apple MPS

Cloud Platforms: AWS, EC2, S3, Lambda, SageMaker